WORKED SOLUTION

STEP I 2008, Question 3

Pure • Inequalities by Rearrangement

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q3

The key idea

The inequality $(*)$ is just a product of two non-negative factors in disguise. Once we have it, every later inequality is a matter of choosing a, b, c, d so that the two sides of $(*)$ become the two sides we want, while making sure the conditions $c \geq a$ and $d \geq b$ really hold. Because the target inequalities are symmetric, we are free to put the variables in order first, and that ordering is what supplies those conditions.

Solution

Proving $(*)$

Bring everything to one side and factorise:

$$ab + cd - bc - ad = a(b-d) - c(b-d) = (c-a)(d-b).$$

If $c \geq a$ and $d \geq b$, both factors are non-negative, so their product is non-negative. Hence $ab + cd \geq bc + ad$, with equality exactly when $c = a$ or $d = b$.

Part (i)

Two variables. Suppose $x \geq y$. In $(*)$ take

$$a = y, \quad b = y, \quad c = x, \quad d = x.$$

The conditions $c \geq a$ and $d \geq b$ both say $x \geq y$, which is given. Then $ab + cd = y^2 + x^2$ and $bc + ad = xy + xy$, so $(*)$ gives

$$x^2 + y^2 \geq 2xy.$$

Three variables. Now suppose also that $x \geq z$ and $y \geq z$. Take

$$a = z, \quad b = z, \quad c = x, \quad d = y.$$

Here $c \geq a$ is $x \geq z$ and $d \geq b$ is $y \geq z$, both given. Now $ab + cd = z^2 + xy$ and $bc + ad = zx + zy$, so

$$z^2 + xy \geq xz + yz.$$

Adding this to $x^2 + y^2 \geq 2xy$ (which we may use, since $x \geq y$):

$$x^2 + y^2 + z^2 + xy \geq 2xy + xz + yz,$$

and subtracting xy from both sides gives

$$x^2 + y^2 + z^2 \geq xy + yz + zx.$$

So far this is proved only when $x \geq y \geq z$.

Ask yourself How can a result proved only for $x \geq y \geq z$ give one for all x, y, z?

Both sides of $x^2 + y^2 + z^2 \geq xy + yz + zx$ are symmetric: swapping any two of the variables leaves each side unchanged. Given any real x, y, z, list them in decreasing order as $X \geq Y \geq Z$. By what we have just shown, $X^2 + Y^2 + Z^2 \geq XY + YZ + ZX$. But this is the same inequality as the one for x, y, z, only with the terms in a different order. So the inequality holds for all real x, y and z.

(Equality needs equality in both inequalities we added, which forces $x = y = z$.)

Part (ii)

The expression is the sum of three pairs, each a number plus its reciprocal:

$$\left(\frac{s}{t} + \frac{t}{s}\right) + \left(\frac{t}{r} + \frac{r}{t}\right) + \left(\frac{r}{s} + \frac{s}{r}\right).$$

So it is enough to show that $\dfrac{p}{q} + \dfrac{q}{p} \geq 2$ for any positive p, q, the analogue of $x^2 + y^2 \geq 2xy$.

The expression $\frac{p}{q} + \frac{q}{p}$ is symmetric in p and q, so we may label them so that $p \geq q > 0$. In $(*)$ take

$$a = q, \quad b = \frac{1}{p}, \quad c = p, \quad d = \frac{1}{q}.$$

Then $c \geq a$ is $p \geq q$, and $d \geq b$ is $\frac{1}{q} \geq \frac{1}{p}$, which also follows from $p \geq q$ because p and q are positive. Now

$$ab + cd = \frac{q}{p} + \frac{p}{q}, \qquad bc + ad = \frac{p}{p} + \frac{q}{q} = 2,$$

so $(*)$ gives $\dfrac{p}{q} + \dfrac{q}{p} \geq 2$.

Applying this to the pairs (s,t), (t,r) and (r,s) and adding the three results,

$$\frac{s}{t} + \frac{t}{r} + \frac{r}{s} + \frac{t}{s} + \frac{r}{t} + \frac{s}{r} \geq 2 + 2 + 2 = 6$$

for all positive r, s and t, as required. Equality holds exactly when $r = s = t$, since each pair needs its two numbers equal.

Checks

- Try $x = 3$, $y = 1$, $z = -2$ in part (i): the left side is 14 and the right side is $3 - 2 - 6 = -5$. Both sides are unchanged if we permute the three values, as the symmetry argument needs.
- An independent proof of part (i): $2(x^2 + y^2 + z^2 - xy - yz - zx) = (x - y)^2 + (y - z)^2 + (z - x)^2 \geq 0$.
- With $r = 2$, $s = 1$, $t = 1$ the expression in part (ii) is $1 + \frac{1}{2} + 2 + 1 + 2 + \frac{1}{2} = 7 \geq 6$. A computer checking every step on 20,000 random exact rational cases found no failures, and values such as $r = 1.001$, $s = 1$, $t = 0.999$ give a sum just above 6.

Common slips

- Using $(*)$ without checking $c \geq a$ and $d \geq b$. With the wrong assignment the inequality can point the other way.
- In part (ii), forgetting that $\frac{1}{q} \geq \frac{1}{p}$ uses positivity: for negative numbers taking reciprocals does not reverse the order in the same way.
- Proving part (i) only for $x \geq y \geq z$ and not explaining why symmetry gives the general case.

Beyond STEP

The inequality $(*)$ is the two-term case of the *rearrangement inequality*: if two lists are sorted, the sum of products is largest when they are paired in the same order and smallest when paired in opposite orders. Both results here, and AM–GM for two terms, drop out of it, and it is closely related to Chebyshev's sum inequality. Siklos, *Advanced Problems in Mathematics*, has more STEP-style inequality practice, and Bartle and Sherbert, *Introduction to Real Analysis*, shows how such order arguments underpin analysis.

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